

PomoMate — Deployment Guide

Overview

PomoMate is a collaborative Pomodoro timer app with real-time P2P rooms, supporting up to 8 users simultaneously on Android, iOS, and Web.

Architecture:

- React Native mobile app (Expo)
 - Node.js backend (Express + WebSocket)
 - Supabase (PostgreSQL + Storage + Auth)
 - WebRTC P2P (mesh topology, host-authoritative)
 - AdMob (free tier)
 - RevenueCat (premium subscriptions)
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Prerequisites

Development Environment

- Node.js 20+
- npm 10+
- Git
- Expo CLI
- Android Studio (for Android builds)
- Xcode (for iOS builds, macOS only)

Production Environment

- Linux server (Abacus SuperComputer recommended)
- PostgreSQL 15
- Nginx
- PM2
- Certbot (for SSL)
- Docker & Docker Compose (optional)

Third-Party Services

1. Supabase

- Self-hosted or cloud instance
- Database + Storage + Auth configured

2. RevenueCat

- Account created
- Products configured (monthly, yearly premium)
- API keys obtained

3. AdMob

- Google AdMob account

- App registered
- Ad unit IDs created

4. Domain & DNS

- Domain purchased
- DNS A records configured

Quick Start (Development)

1. Clone Repository

```
git clone https://github.com/your-org/pomomate.git
cd pomomate
```

2. Install Dependencies

```
# Mobile app
npm install

# Backend
cd server
npm install
cd ..
```

3. Configure Environment

```
# Mobile
cp .env.example .env
nano .env
# Fill in EXPO_PUBLIC_* variables

# Backend
cd server
cp .env.example .env
nano .env
# Fill in Supabase, JWT, CORS variables
cd ..
```

4. Start Development Servers

```
# Terminal 1: Backend
cd server
npm run dev

# Terminal 2: Mobile app
npm start
# Press 'a' for Android, 'i' for iOS, 'w' for Web
```

Production Deployment

Backend Deployment (Abacus SuperComputer)

1. Initial Setup

```
# SSH into server
ssh user@your-server.com

# Clone repository
cd /home/ubuntu
git clone https://github.com/your-org/pomomate.git
cd pomomate

# Install dependencies
cd server
npm ci --only=production
npm run build
cd ..
```

2. Database Setup

```
# Create PostgreSQL database and user
sudo -u postgres psql
CREATE DATABASE pomomate;
CREATE USER pomomate_user WITH PASSWORD 'SECURE_PASSWORD';
GRANT ALL PRIVILEGES ON DATABASE pomomate TO pomomate_user;
\q

# Run migrations
psql -h localhost -U pomomate_user -d pomomate < server/migrations/
001_initial_schema.sql
psql -h localhost -U pomomate_user -d pomomate < server/migrations/
002_rls_policies.sql
psql -h localhost -U pomomate_user -d pomomate < server/migrations/
003_storage_buckets.sql
```

3. Environment Configuration

```
cp .env.production.example .env.production
nano .env.production
# Fill in all production values
```

4. Nginx Setup

```
sudo cp deployment/nginx/pomomate.conf /etc/nginx/sites-available/
sudo ln -s /etc/nginx/sites-available/pomomate.conf /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
```

5. SSL Certificate

```
sudo certbot --nginx -d api.pomomate.app
```

6. Start Services

```
# Install PM2 globally
npm install -g pm2

# Start backend
pm2 start deployment/config/ecosystem.config.js

# Save PM2 state
pm2 save

# Enable PM2 on boot
pm2 startup

# Run the command PM2 outputs
```

7. Setup Automated Backups

```
chmod +x deployment/scripts/backup.sh

# Add to crontab
crontab -e
# Add: 0 2 * * * /home/ubuntu/pomomate/deployment/scripts/backup.sh
```

Mobile App Deployment

Android Build

```
# Configure app.json
nano app.json
# Update version, package name, etc.

# Build APK
eas build --platform android --profile production

# Or build AAB for Google Play
eas build --platform android --profile production:aab
```

iOS Build

```
# Configure app.json
nano app.json
# Update version, bundle identifier, etc.

# Build IPA
eas build --platform ios --profile production
```

Submit to Stores

```
# Google Play Store
eas submit --platform android

# Apple App Store
eas submit --platform ios
```

Configuration

Environment Variables

Mobile App (.env)

```
EXPO_PUBLIC_API_URL=https://api.pomomate.app
EXPO_PUBLIC_SUPABASE_URL=https://your-instance.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
EXPO_PUBLIC_WEBRTC_SIGNALING_URL=wss://api.pomomate.app/ws/signaling
EXPO_PUBLIC_ENV=production
```

Backend (.env.production)

```
PORT=3000
NODE_ENV=production
SUPABASE_URL=https://your-instance.supabase.co
SUPABASE_ANON_KEY=your-anon-key
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key
CORS_ORIGIN=https://pomomate.app
JWT_SECRET=your-supabase-jwt-secret
DB_USER=pomomate_user
DB_PASSWORD=your-secure-password
```

RevenueCat Configuration

1. Update `src/services/monetization/RevenueCatService.ts`
2. Replace API keys with production values
3. Configure products in RevenueCat dashboard

AdMob Configuration

1. Update `src/services/monetization/AdMobService.ts`
2. Replace test ad unit IDs with production IDs
3. Enable ads in AdMob console



Monitoring

Health Checks

```
# Backend health
curl https://api.pomomate.app/health

# PM2 status
pm2 status

# PM2 logs
pm2 logs pomomate-backend --lines 50

# PM2 monitoring
pm2 monit
```

Database Monitoring

```
# Check connections
psql -U pomomate_user -d pomomate -c "SELECT count(*) FROM pg_stat_activity;"

# Check table sizes
psql -U pomomate_user -d pomomate -c "SELECT schemaname, tablename,
pg_total_relation_size(schemaname||'.'||tablename) AS size FROM pg_tables ORDER BY
size DESC LIMIT 10;"
```

Nginx Logs

```
# Access logs
sudo tail -f /var/log/nginx/access.log

# Error logs
sudo tail -f /var/log/nginx/error.log
```



Updates & Maintenance

Deploy New Version

```
cd /home/ubuntu/pomomate
git pull origin main
./deployment/scripts/deploy.sh
```

Database Migrations

```
# Create new migration
nano server/migrations/004_your_migration.sql

# Apply migration
psql -U pomomate_user -d pomomate < server/migrations/004_your_migration.sql
```

Rollback

```
# Rollback PM2 to previous version
pm2 reload pomomate-backend --update-env

# Or restore from backup
gunzip -c /home/ubuntu/pomomate/backups/pomomate_backup_YYYYMMDD_HHMMSS.sql.gz | psql
-U pomomate_user pomomate
```



Security

Checklist

- [x] SSL/TLS enabled
- [x] JWT secrets secure

- [x] Database passwords strong
- [x] RLS policies enforced
- [x] CORS configured
- [x] Rate limiting enabled
- [x] Secrets not committed to git

Security Updates

```
# Update packages
npm audit
npm audit fix

# Update system
sudo apt update && sudo apt upgrade -y
```

Troubleshooting

Backend Won't Start

```
# Check logs
pm2 logs pomomate-backend

# Check port conflicts
sudo lsof -i :3000

# Restart services
pm2 restart pomomate-backend
```

Database Connection Failed

```
# Check PostgreSQL status
sudo systemctl status postgresql

# Test connection
psql -U pomomate_user -d pomomate -c "SELECT 1;"
```

WebSocket Not Connecting

```
# Check Nginx WebSocket config
sudo nginx -t

# Check firewall
sudo ufw status
```

Mobile App Build Failed

```
# Clear cache
expo start -c

# Clear node_modules
rm -rf node_modules
npm install
```

Support

- **Documentation:** </docs>
- **GitHub Issues:** [\[Repository Issues\]](#)
- **Email:** dev@pomomate.app

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